

Assessment Specification – Portfolio Task 3

Module Details

Module Code	UFCE8B-30-3
Module Title	Data Science for Cyber Security
Module Leader	Ahsan Ikram
Module Tutors	Ahsan Ikram, Andrew McCarthy
Year	2025-26
Task	03
Total number of assessments for this module	01
Weighting	20%
Date issued to students	16/02/2026

Task Specification

You are a cyber security analyst consulting for a private company. They have recently been under various phishing attacks. You have quarantined a collection of sample files from prospective phishing emails and links. You are planning to investigate the files further using Python and data science related to malware analysis.

Sample Files: File1.task3, File 2.task3, File 3.task3, dataset_task3.csv and readme.txt (Available on Blackboard.)

Files description: All the files are in the Windows Portable Executable (PE) format.

Deliverable: You are required to use Jupyter notebook to complete this task.

Analyse all the files to perform the following tasks,

	Task	Marks
1	Print all the Section headers.	2
2	Print all the imported libraries.	2
3	Print a summary of all the string features of length four or more characters. For example, total count, average length, frequency etc.	4
4	Create a list of following suspicious imports ('GetProcAddress' 'CreateRemoteThread', 'TerminateProcess', 'VirtualAlloc', and 'WriteProcessMemory') and write a script to check if the calls exist in a sample file. If a file has more than 50% matches, it should be considered malicious.	4
5	Using the training data in the dataset_task3.csv file, - Train and test a Decision Tree and Neural Network model.	8

	- Compare the accuracy of the two models. <i>Note: The readme.txt file contains brief descriptions of the features.</i>	
	Total	20